Loss function, cost/objective function

- Given ground truth labels (y_i), scores $f(x_i, \mathbf{W})$
  - how unhappy are we with the scores?
- Loss function or objective/cost function measures unhappiness

- During training, **want to find the parameters W that minimize the loss function**

Simpler example: binary classification

- Two classes (e.g., “cat” and “not cat”)
  - AKA “positive” and “negative” classes





cat

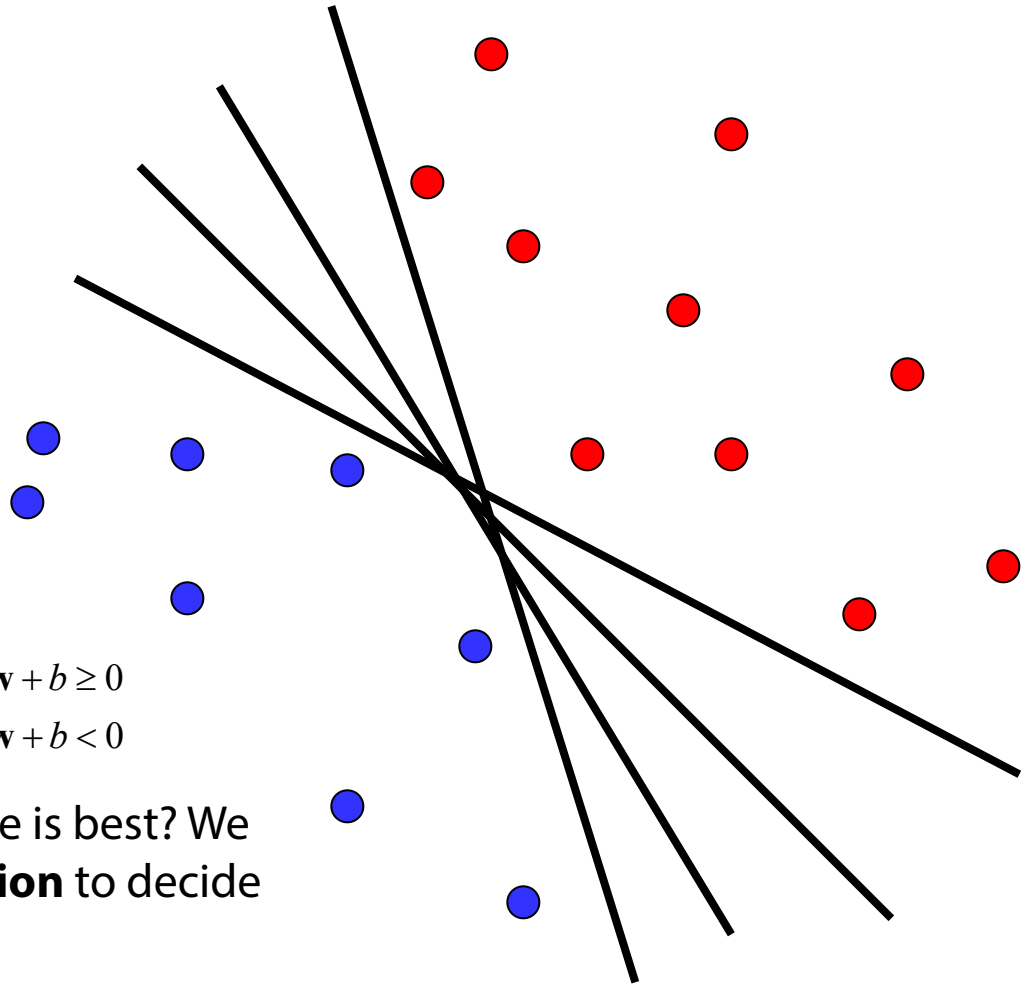




not cat

Linear classifiers

- Find linear function (*hyperplane*) to separate positive and negative examples

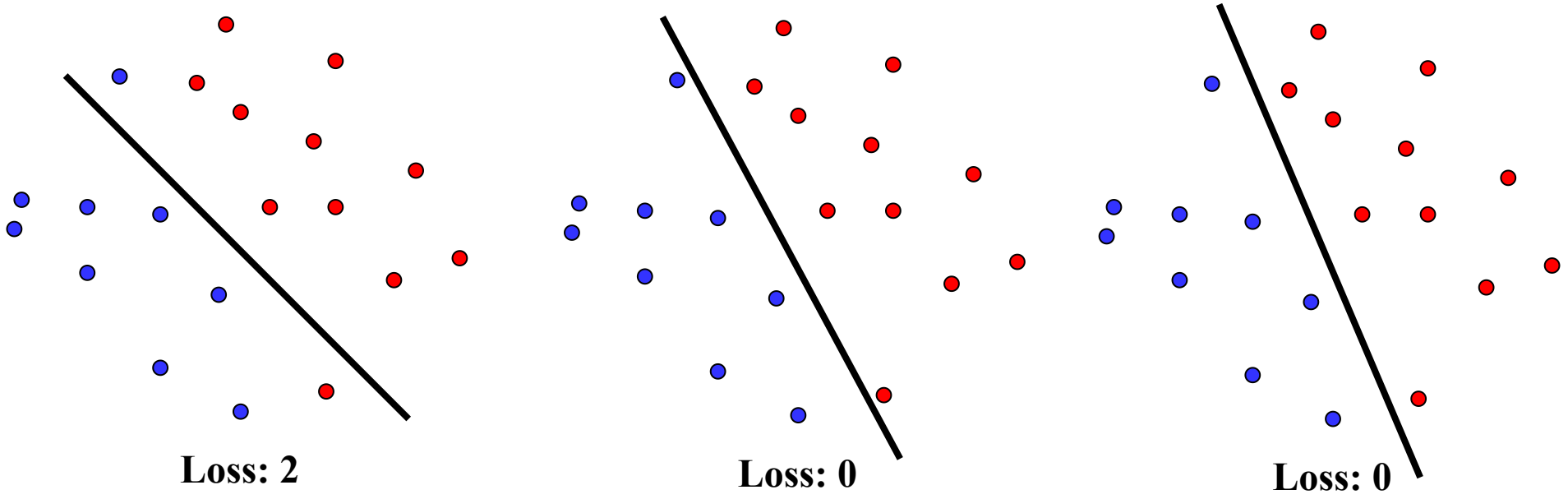


$$\mathbf{x}_i \text{ positive}: \quad \mathbf{x}_i \cdot \mathbf{w} + b \geq 0$$
$$\mathbf{x}_i \text{ negative}: \quad \mathbf{x}_i \cdot \mathbf{w} + b < 0$$

Which hyperplane is best? We need a **loss function** to decide

What is a good loss function?

- One possibility: Number of misclassified examples
  - Problems: discrete, can't break ties
  - We want the loss to lead to *good generalization*
  - We want the loss to work for more than 2 classes



Softmax classifier

- Interpret Scores as unnormalized log probabilities of classes

$$f(x_i, W) = Wx_i \quad \text{(score function)}$$

$$\frac{e^{f_{y_i}}}{\sum_j e^{f_j}}$$ softmax function

Squashes values into *probabilities* ranging from 0 to 1

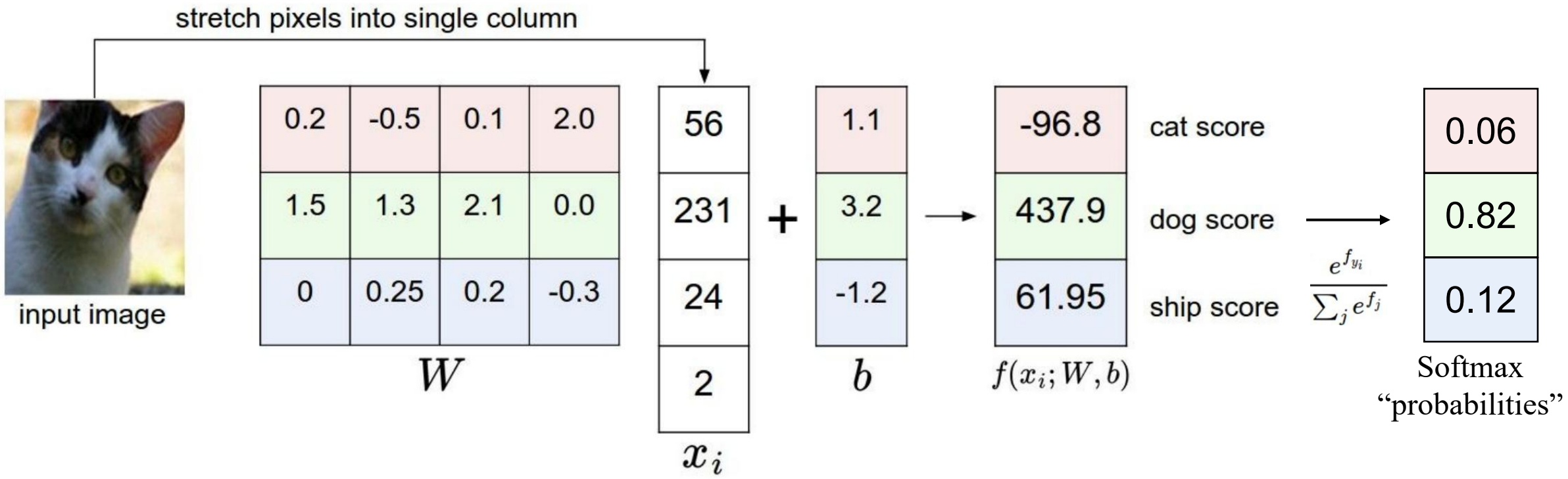
$$P(y_i \mid x_i; W)$$

Example with three classes:

$$[1, -2, 0] \rightarrow [e^1, e^{-2}, e^0] = [2.71, 0.14, 1] \rightarrow [0.7, 0.04, 0.26]$$

Softmax classifier

Example with an image with 4 pixels, and 3 classes (cat/dog/ship)



Cross-entropy loss

$$f(x_i, W) = Wx_i$$ (score function)

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f_{y_i} : score of correct class

$$L_i = -\log\left(\frac{e^{f_{y_i}}}{\sum_j e^{f_j}}\right)$$

$$L_i = -f_{y_i} + \log\sum_j e^{f_j}$$

We call L_i *cross-entropy loss*